

Studying the Effect of Physical Realism on Time Perception in a HAZMAT VR Simulation

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Figure 1: Left to right. The experiment corridor, virtual experiment corridor, physical gas monitor and vent props, and the virtual gas monitor.

ABSTRACT

Our research focuses on how physical props in virtual reality (VR) can affect users' time perception. We designed an experiment with the goal of comparing users' perception of time when using physical props in VR as compared to standard controllers and only virtual elements. In order to quantify this effect, time estimates for both conditions are compared to time estimates for a matching real-world task. In this experiment, participants assume the role of a firefighter trainee, going through a HAZMAT scenario, where they touch and interact with different physical props that match the virtual elements of the scene.

Index Terms: Human-centered computing—Time-perception—Virtual Reality; Human-centered computing—Human computer interaction (HCI)—Interaction devices—Haptic devices.

1 INTRODUCTION

HAZMAT (hazardous material) technicians often need to respond to emergencies where their perception of time is essential for the success of the operation. One time-critical situation is monitoring for hazardous gases. The accurate and precise identification of leaks is imperative to the safety of those around. A gas monitor tool helps HAZMAT technicians detect and identify hazardous gases. To achieve accurate results, this device must be held close to the source of the hazardous gas for a sufficient amount of time. To identify and accurately display gas levels, HAZMAT workers are advised to hold the device at the area of interest for 30 seconds.

Compared to traditional training, virtual reality (VR) training has recently started to be widely considered and tested as a high-fidelity low-cost training alternative. Entire rooms and even buildings can be virtually crafted to train HAZMAT technicians, whereas, with

traditional training, these commodities are costly or impossible to access. A VR training program typically utilizes a single VR headset and two hand-held controllers per trainee. These devices limit the user's experience in terms of what they can feel. The hand-held controllers are used to represent the hands of the user. The user can interact with virtual objects by operating these controllers, however, as everything other than the controllers and headset are virtual, users are not able to precisely feel the form and weight of these objects. Thus, the physical experience is limited.

One method of overcoming this limitation is by creating physical props tracked in VR the same way standard controllers are. Users would be able to interact with the physical devices as they would interact with the respective real-world objects: by physically holding them, feeling their shape, weight, and texture.

Previous work has demonstrated a *time compression* effect in VR, where the elapsed time is perceived as being shorter than the actual elapsed time due to factors such as reduced bodily awareness and engagement [11, 12]. Our aim is to determine whether the physical realism of VR props and hand representation can influence time perception. To that end, we present the design of an experiment where participants perform a time estimation task in the real world, VR with low physical realism, VR with high physical realism.

In our proposed protocol, when physical props are used, the participants hold and operate a 3D printed device that matches the weight and form of a real gas monitor (Figure 1, center right), and navigate through a corridor of physical props where they touch and feel objects with their hands. Participants then mentally count the time between two signals. We compare the seconds a participant estimates in each session in order to see how much their perception of time has changed, if any relative to the control real-world condition.

2 PASSIVE HAPTICS

Passive haptics provide a sense of physical touch in VR by creating real-world physical objects that match the shape and pose of virtual objects [8]. Researchers have demonstrated the use of such physical objects and tangible interaction to enhance 3D virtual interactions.

A notable aspect of passive haptics is that once a user realizes those objects are real, they assume virtual objects they do not touch are also real and that they would physically feel them if they did

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touch them [6]. Because of this, only adjacent objects that the user is likely to touch need to be physically represented.

Using passive haptics with VR improves presence, cognitive mapping, spatial knowledge training, and performance [5]. Hinckley et al. [4] used a physical plastic doll head as a proxy for rotational control of 3D models. As another example, Meehan et al. [10] had users walk on a plank of wood on the floor to provide the tactile feeling of the feet extending over the edge into a deep pit. In other work, Lok et al. [9] presented an approach for allowing virtual objects to dynamically interact with physical props. More recently, Englmeier et al. [2] tracked two differently sized acrylic glass spheres by placing a Vive Puck inside each sphere, and compared handling the fully tangible spheres to a closely related controller-as-proxy interaction technique.

Utilizing passive haptics does not necessarily result in better performance, and that should not necessarily be the goal. Batmaz et al. [1] found that adding a passive-haptic soft-surface wall increased time to hitting that wall (compared to a purely virtual wall), possibly due to concerns that reaching too quickly for the wall could hurt (like it could in the real world). For training of real-life incidents, a goal of passive haptics might be to better match real-world performance instead of trying to do better than the real world by using magical interaction techniques.

Based on these findings and on initial input from firefighters, we expect that the use of passive haptics will result in better preparing firefighters for the real-world incidents they encounter.

3 DESIGN OF THE EXPERIMENT

Traditional HAZMAT training does not allow for the experiential use of a gas monitor and critical incidents that require the use of such a device are quite rare. Thus, informed by input from firefighters, we identified HAZMAT training as an ideal use case for VR. We built different versions of gas monitors including 1) a custom steamVR-compatible controller that matches a real gas monitor [3] and 2) a real gas monitor with an attached an HTC Vive tracking puck [7]. For this experiment, we simplified the hardware by 3D printing a shell matching the gas monitor with an HTC Vive Puck attached.

A critical requirement for a gas monitor to accurately detect dangerous gases is to hold the device in suspected areas for a sufficient period of time. If time distortion would occur in a HAZMAT VR training module, then trainees might not hold the device for the necessary time during real-world incidents. Thus we designed an experiment to determine if physical props within VR can help better match perception of time in VR to the real world.

We hypothesize that time estimates in VR more closely match real world estimates when physical props are used as compared to a completely virtual implementation.

3.1 Participant Tasks

We have designed a within-subjects experiment with two experimental conditions and one baseline condition: non-VR real-world with physical props that serves as a baseline; standard VR with virtual objects; and VR with physical props.

We closely matched the virtual environment to the real-world environment, and the tasks across all three conditions (see Figure 1). In the standard VR condition, the physical props are removed from the experiment area. The participant can still see these objects and interact through standard VR controllers; however, no passive haptics are present as the objects are only virtual. In the physical props condition, the participant can both see and feel objects with the utilization of a Leap Motion hand tracking and physical props, and the need for standard controllers is eliminated.

For each condition, the task starts by having the participant pick up the gas monitor directly next to them on the table. Then, they walk to the door and push it open. They walk around the desk, to a vent where they have been told to measure for gases. They hold

the gas monitor device close to the vent as to detect the hazardous gas concentration. Once the device is close enough to the vent, a high pitch sound is played, which signals the start of the timer. The participant starts mentally counting time until they hear a second high pitch sound, at which time they stop counting and report to the experimenter how much time they estimated. In the end, the participant lowers the gas monitor away from the vent and walks back to the starting position.

3.2 Equipment

Figure 1 shows the experimental setup. We created a one-to-one model of the experiment room and used it in the Unity Game Engine. We are using the HTC Vive Pro headset and controllers. The headset is equipped with a wireless adapter to eliminate cables connected to the computer. A Leap Motion Controller is mounted in front of the headset to track hands (for the physical props condition). A door, which has an attached HTC Vive Tracker, is used as a physical prop and can be interacted with by the participant. A table and desk are also used as physical props in the experiment. A 3D printed gas monitor matching the shape and weight of a real gas monitor is held and carried by participants as a physical prop.

3.3 Next Steps

We have started to collect preliminary data via pilots and plan to collect data with 30 participants. We also plan to conduct a follow-up study based on the results of this first study.

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